# **Topic 12**

Correlations and experiments

# Overview

- Correlational research
- The basics of experimenting
- Experimental research

# Correlational research

- Correlational research → association claims
  - "People who often watch violence on TV are more violent than others"
  - "Mental disorders are more common among unemployed people than among people with a job"
  - "Couples who meet online have better marriages“
  - “In countries with a high number of divorces, people generally have a high life expectancy”

# Correlational research

- Correlation coefficient *(r)*

C14 =CORREL(B2:B12,C2:C12)

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Name** | **Height (cm)** | **Weight (lbs)** | | |
| 2 | Barbara | 166 | 172 | | |
| 3 | Melinda | 190 | 242 | | |
| 4 | Fred | 171 | 172 | | |
| 5 | Pamela | 189 | 211 | | |
| 6 | Jacqueline | 178 | 195 | | |
| 7 | Jose | 189 | 278 | | |
| 8 | Rosario | 188 | 271 | | |
| 9 | Rolando | 173 | 197 | | |
| 10 | Orlando | 186 | 194 | | |
| 11 | Deborah | 160 | 238 | | |
| 12 | Minnie | 172 | 182 | | |
| 13 | | | | | |
| 14 | | | 0.513145861 | | |

- Quantify the relationship

- Product-moment correlation coefficient or Pearson's *r*

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

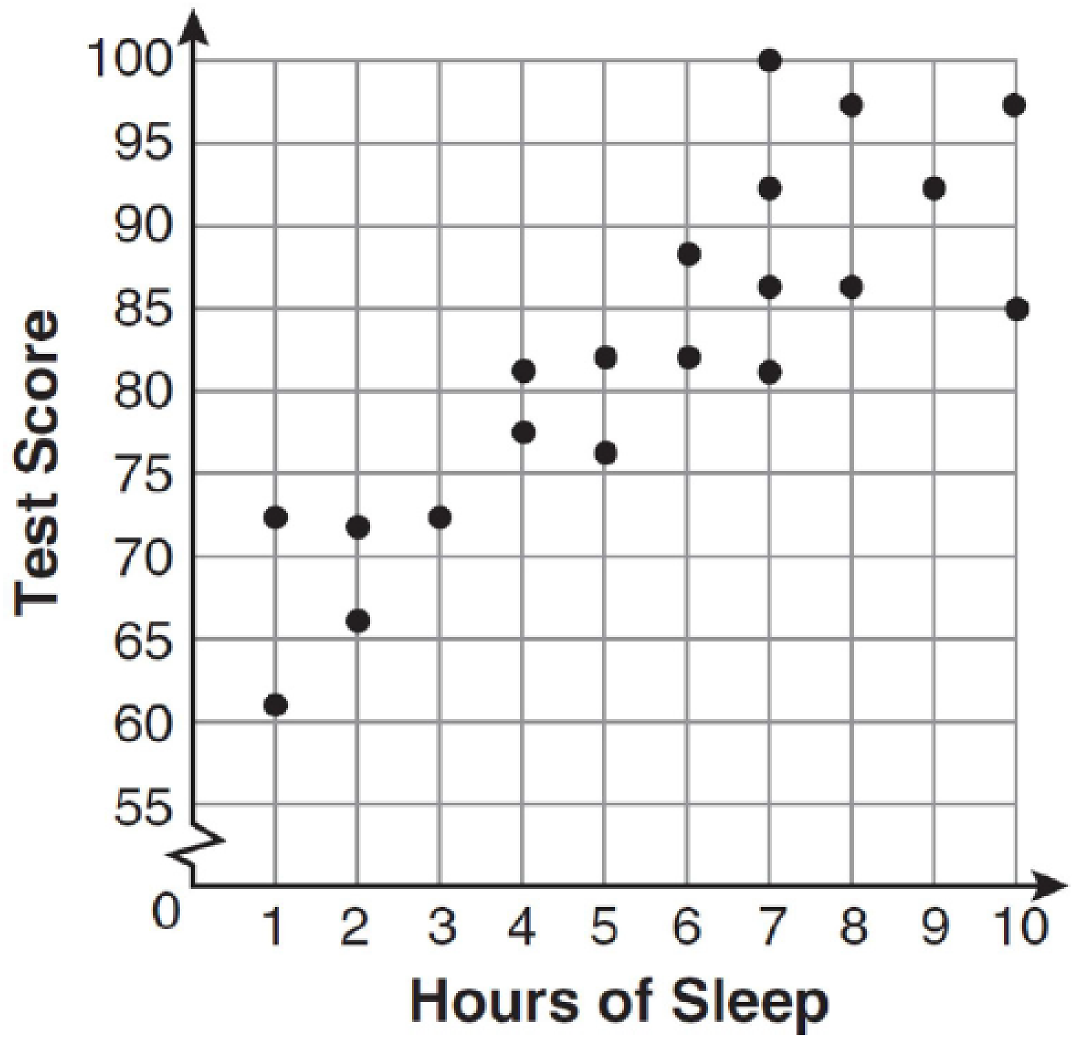
Karl Pearson (1857-1936)

(formula not on exam)

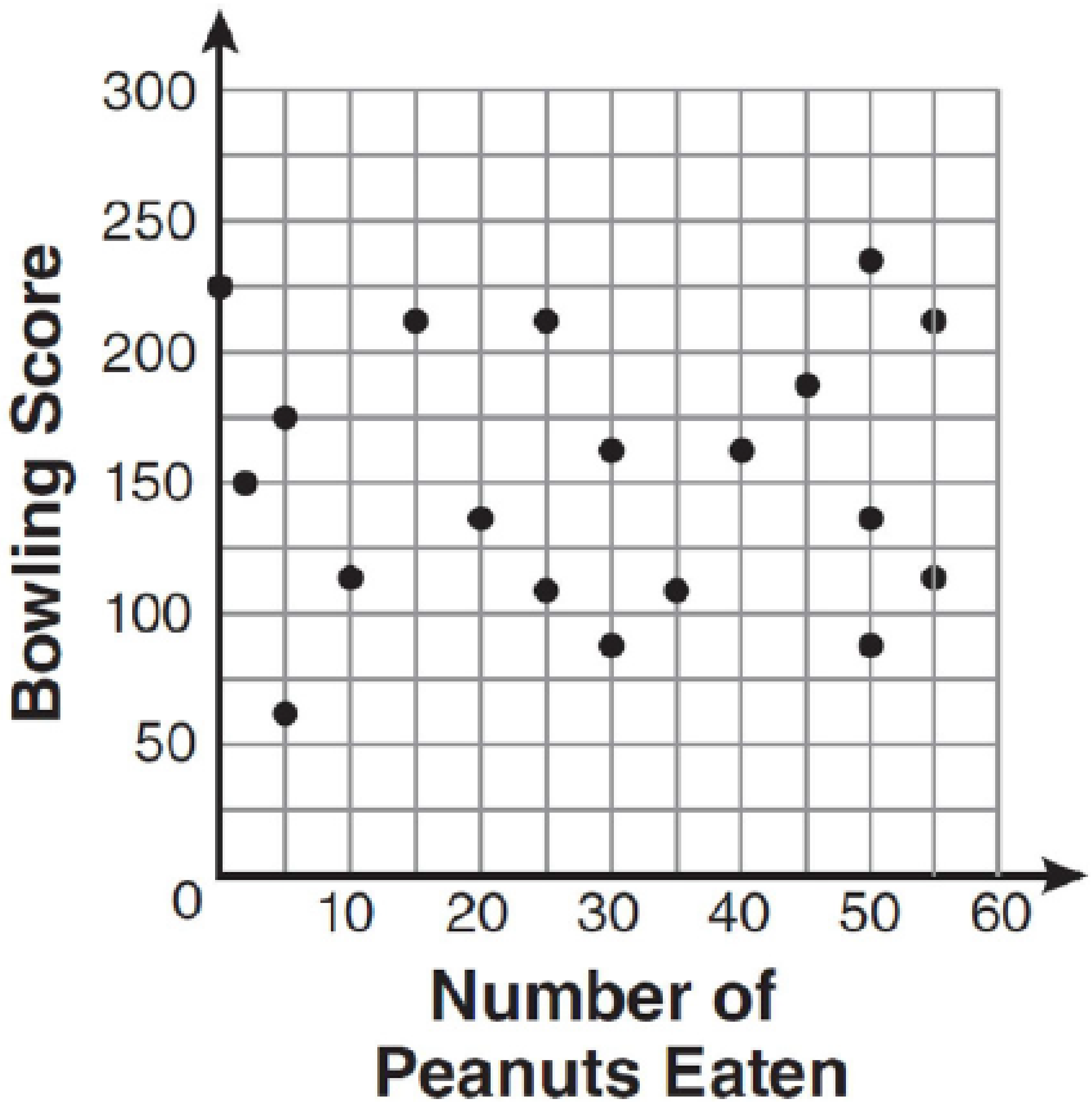
# Correlational research

- Correlation coefficient *(r)*
  - a measure indicating the degree of linear relation between two variables
  - varies between -1.00 and +1.00
    - absolute value indicates the *strength* of the relation
    - sign indicates the *direction* of the relation
  - scatter diagram

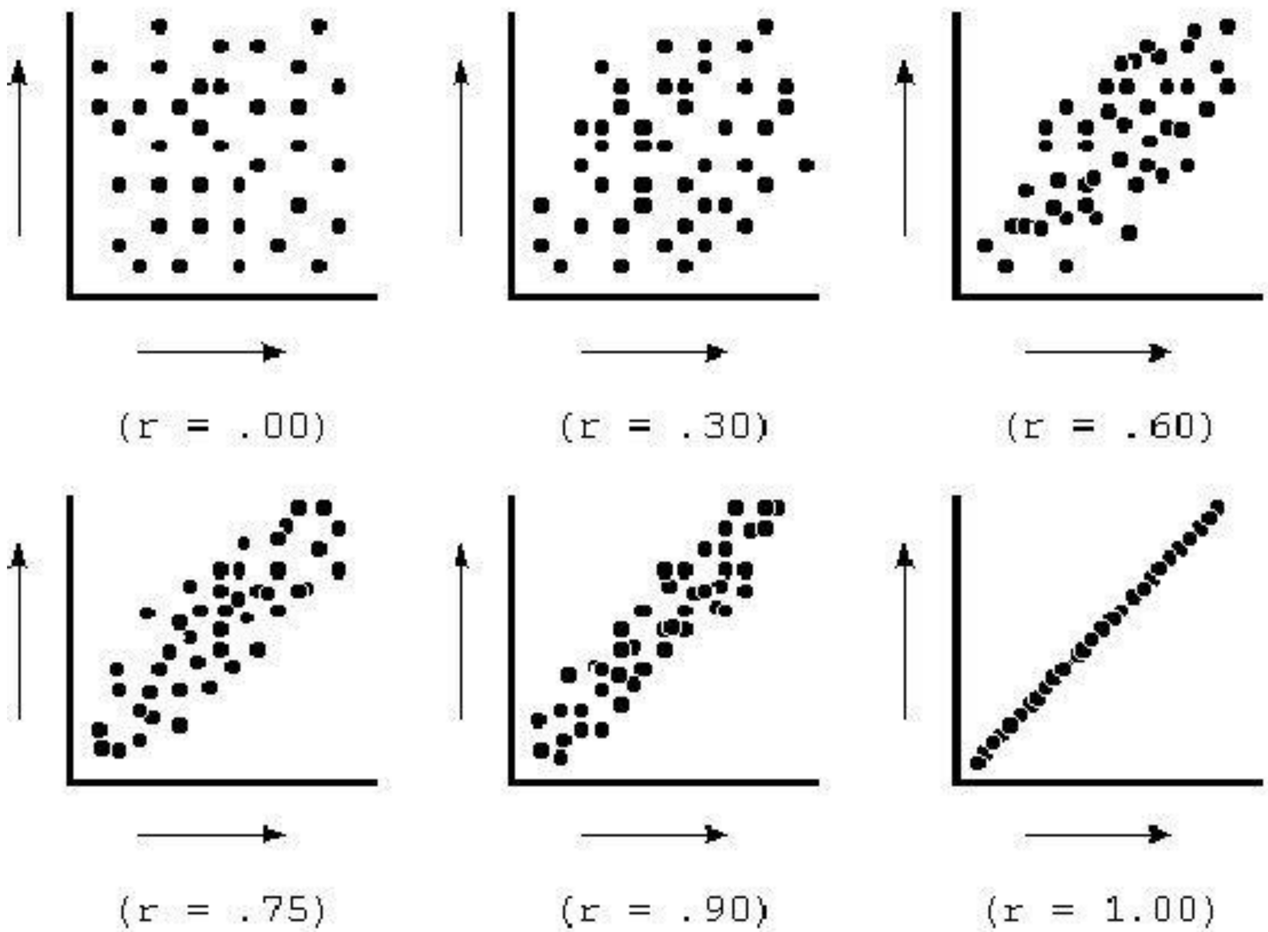
# Correlational research



# Correlational research



# Correlational research



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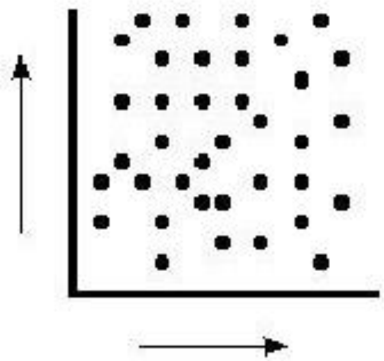
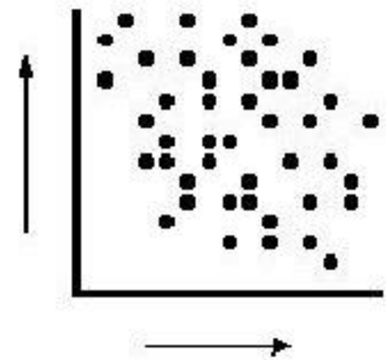
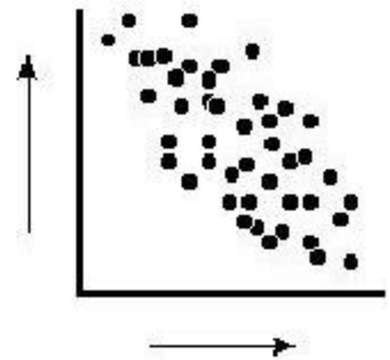


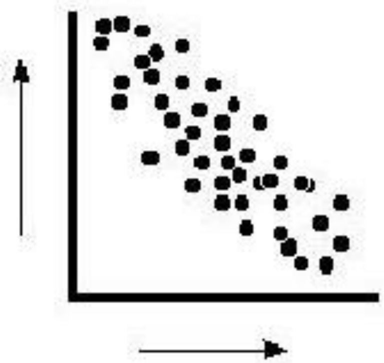
(r = .00)



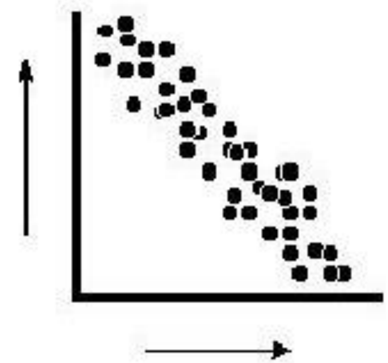
(r = -.30)



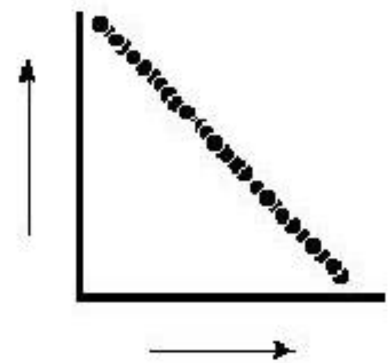
(r = -.60)



(r = -.75)



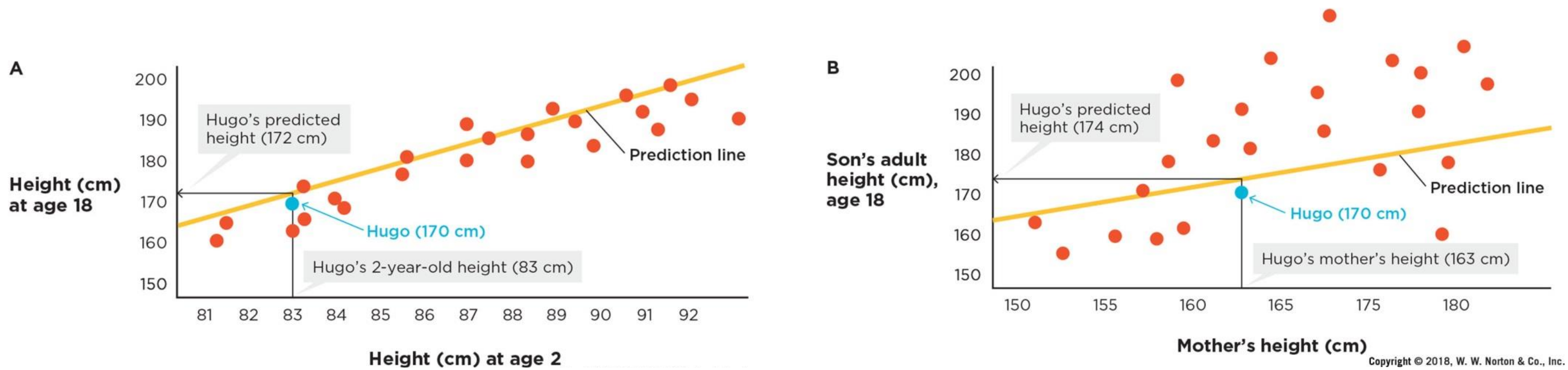
(r = -.90)



(r = -1.00)

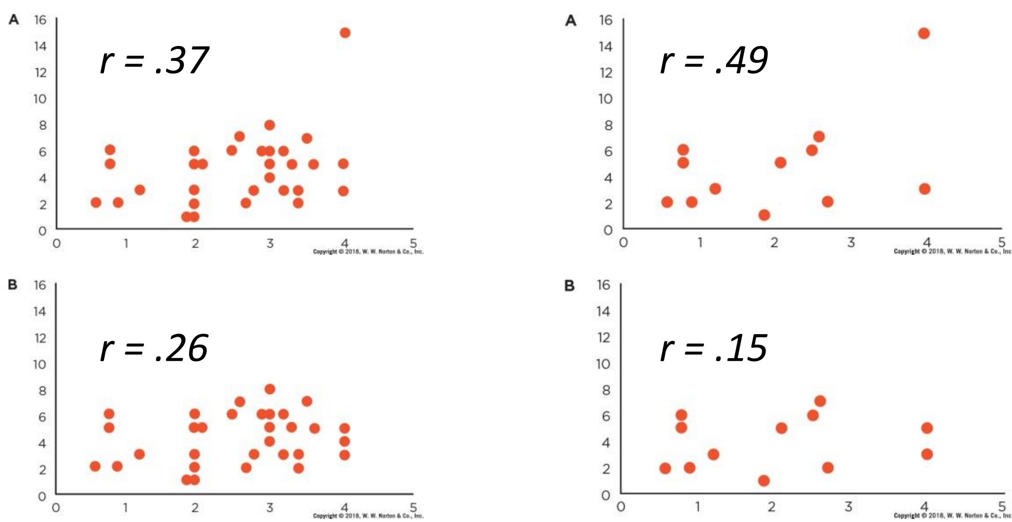
# Correlational research

- Correlational research → association claims
  - Statistical validity: do the data support the claim?
    - What is the strength of the correlation (effect size)?



# Correlational research

- Correlational research → association claims
  - Statistical validity: do the data support the claim?
    - What is the strength of the correlation (effect size)?
    - Could outliers be affecting the correlation?



# Correlational research

- Correlational research → association claims
  - Statistical validity: do the data support the claim?
    - What is the strength of the correlation (effect size)?
    - Could outliers be affecting the correlation?
    - Is the correlation statistically significant?

$$t_{obs} = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

a significance level of 5%, $\alpha = 0.05$

Reject the null hypothesis. $p < 0.05$ (or 0.01, or, more common now 0.001, whichever cutoff point you choose)

DO NOT REJECT the null hypothesis. $p > 0.05$ (or 0.01, or, more common now 0.001, whichever cutoff point you choose)

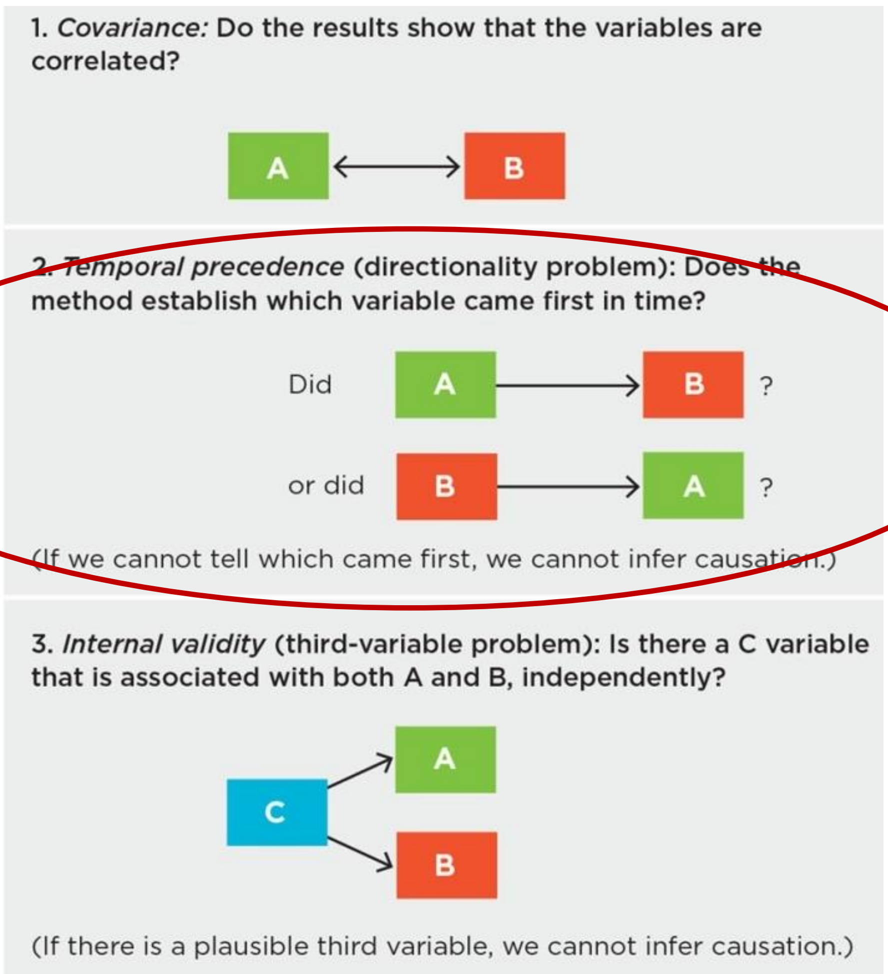
# Correlational research

- There exists a positive correlation between
  - hours people do sports per week
  - general health

  - What does this imply?
    - sports -> health
    - health -> sports
  - Unknown direction of causation (directionality problem)

# Correlational research

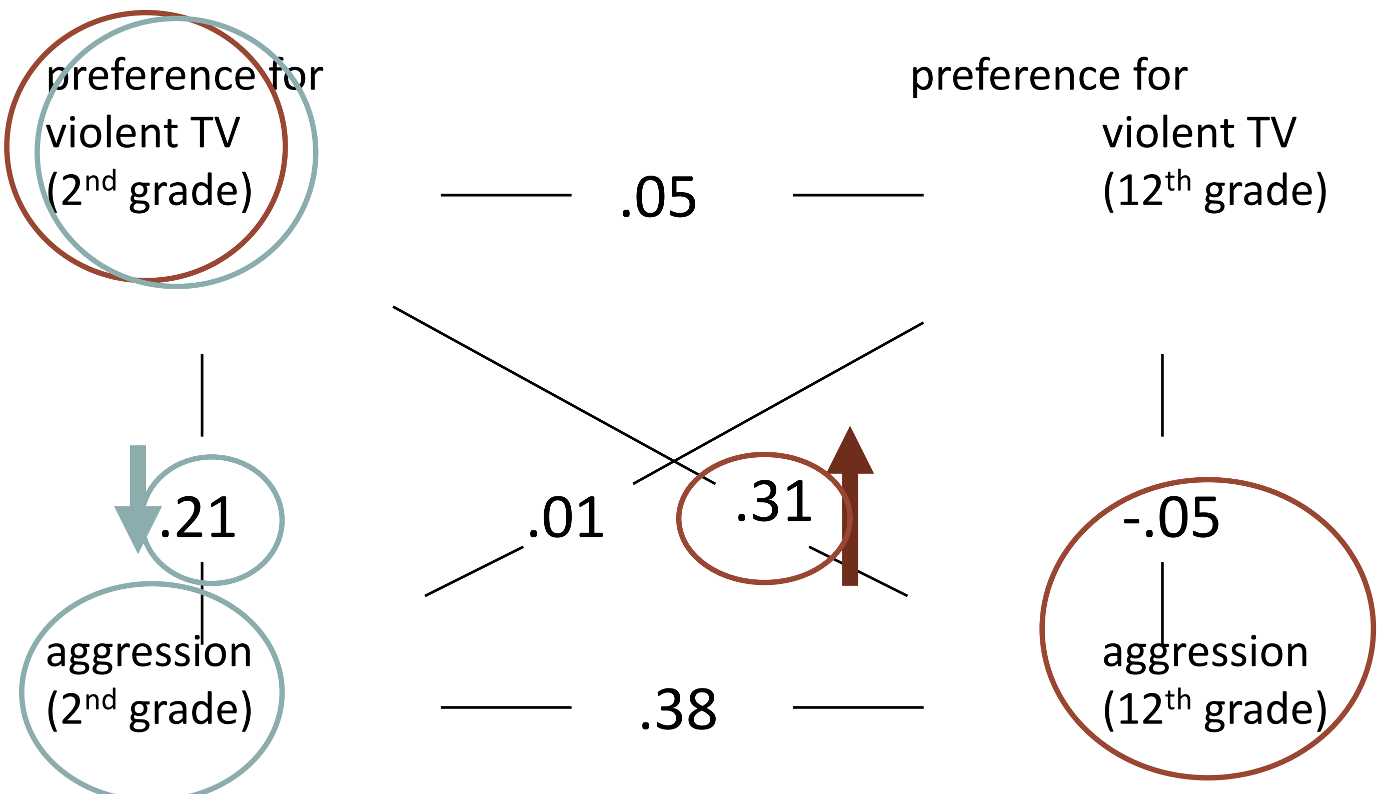
- What are the criteria required to establish causation?




- ***directionality problem*** Sometimes researchers attempt to establish temporal precedence based on correlational research
  → How?

# Correlational research

- by using the cross-lagged-panel procedure
  - A -> B if $r(A_{t1}B_{t2}) > r(A_{t1}B_{t1})$



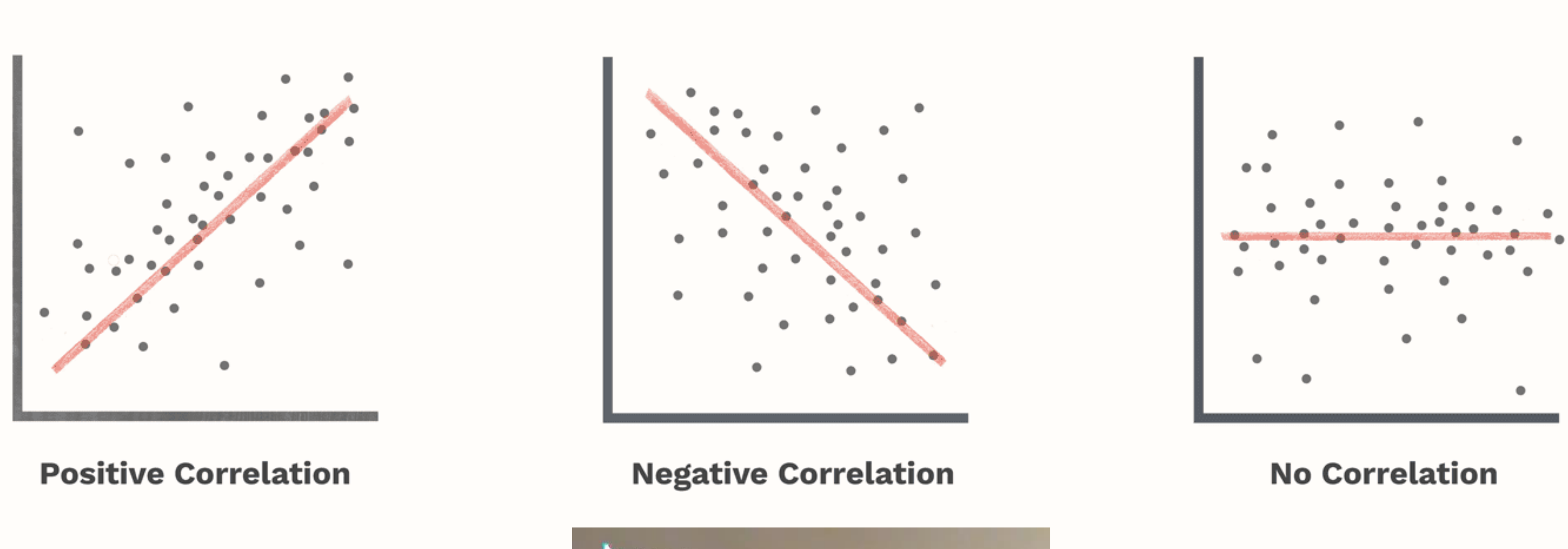

https://www.statisticshowto.com/cross-lagged-panel-design/

# Correlational research

- There exists a positive correlation between
  - the number of ice creams sold per month
  - the number of people drowned per month

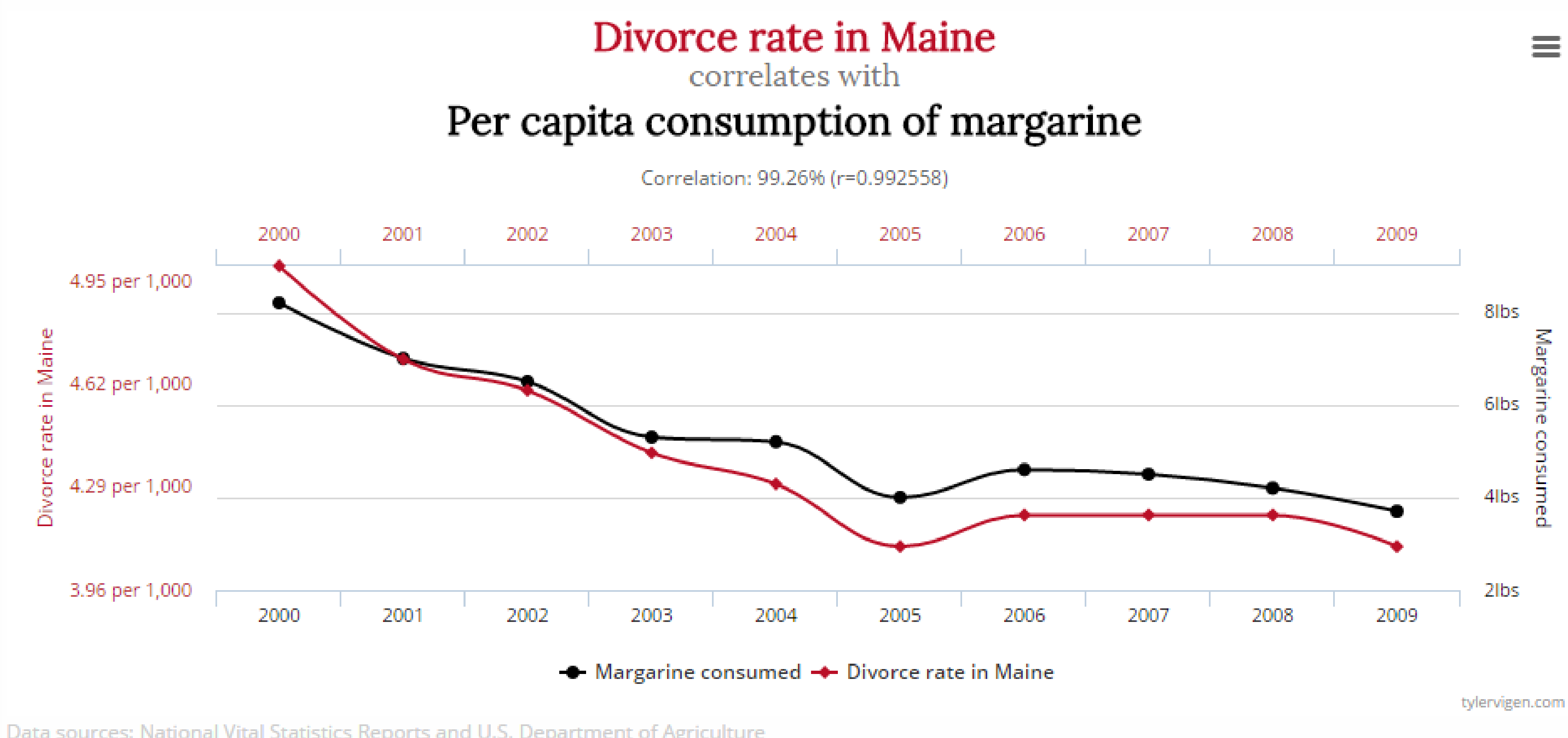
  - What does this imply?
    - ice creams -> drowning
    - drowning -> ice creams
    - High temperature -> more ice creams and drowning
  - 3rd variable problem / Spurious correlation

http://www.psyblog.nl/2014/02/06/correlatie-causaliteitijsjes-piraten-en-de-media/


Positive Correlation
Negative Correlation
No Correlation

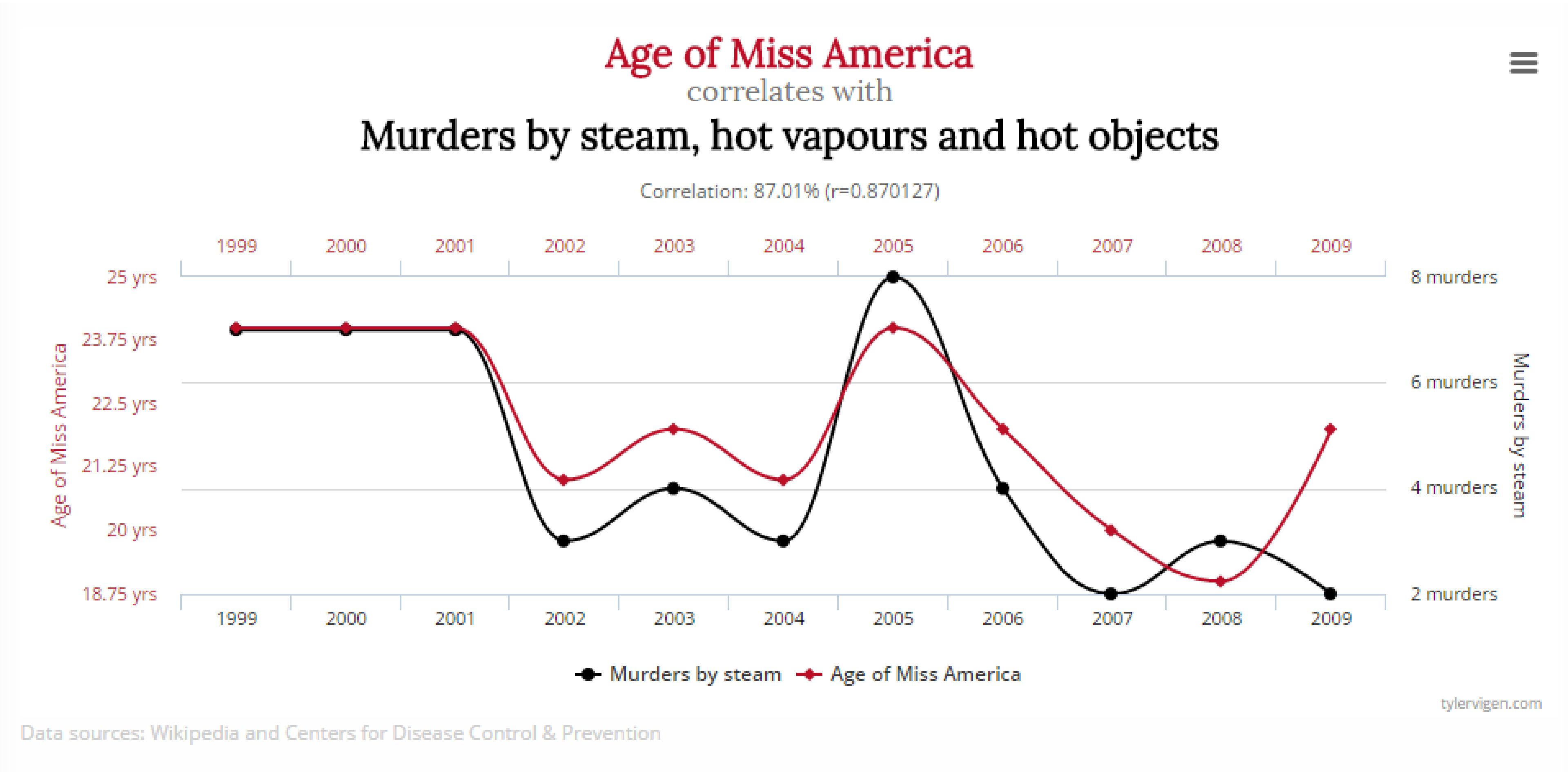

TikTok
@sophiasmithgaler
Correlation does not mean causation... IN GRAPHS

# Correlational research




http://tylervigen.com/spurious-correlations

# Correlational research

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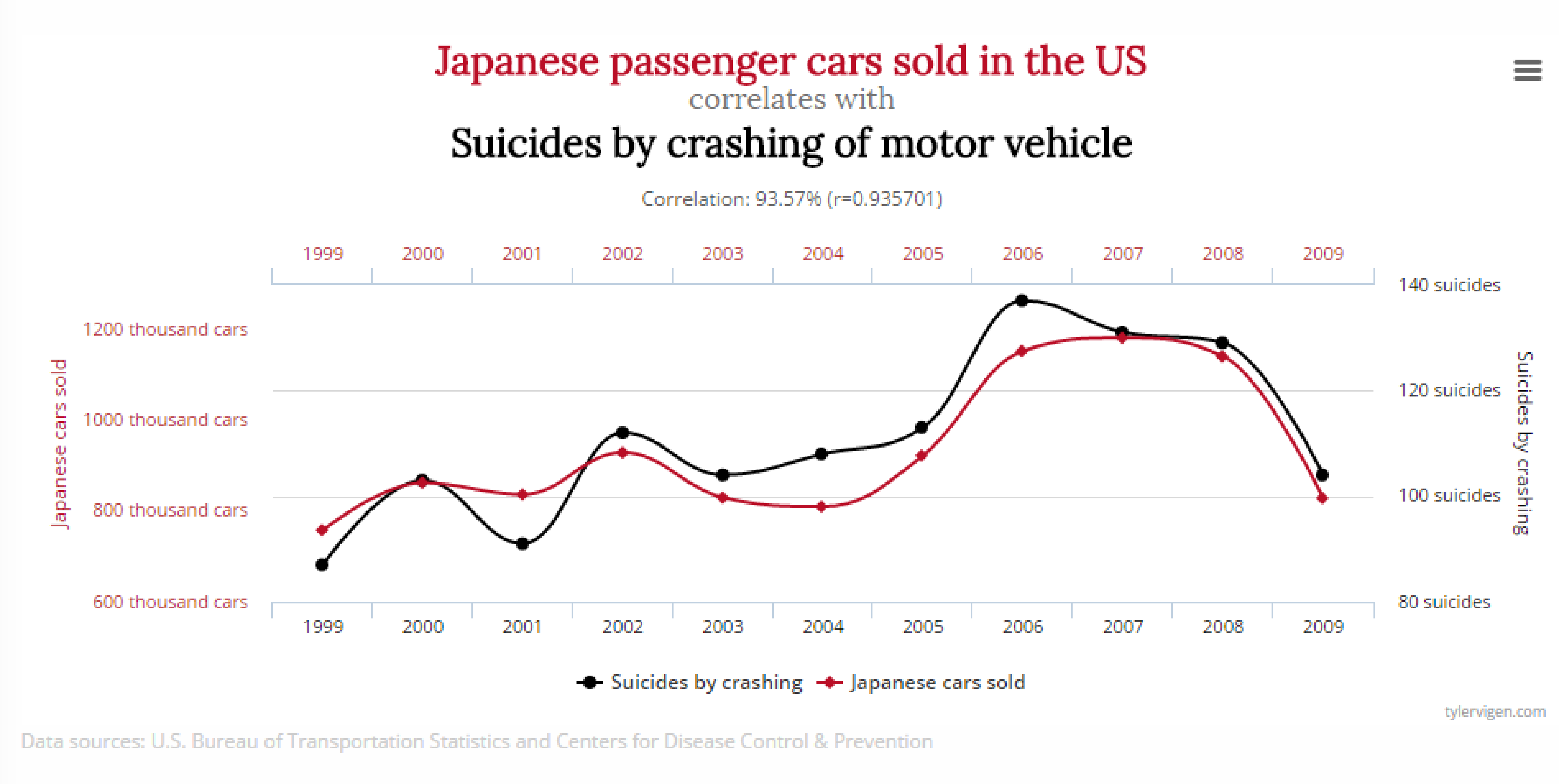
# Correlational research

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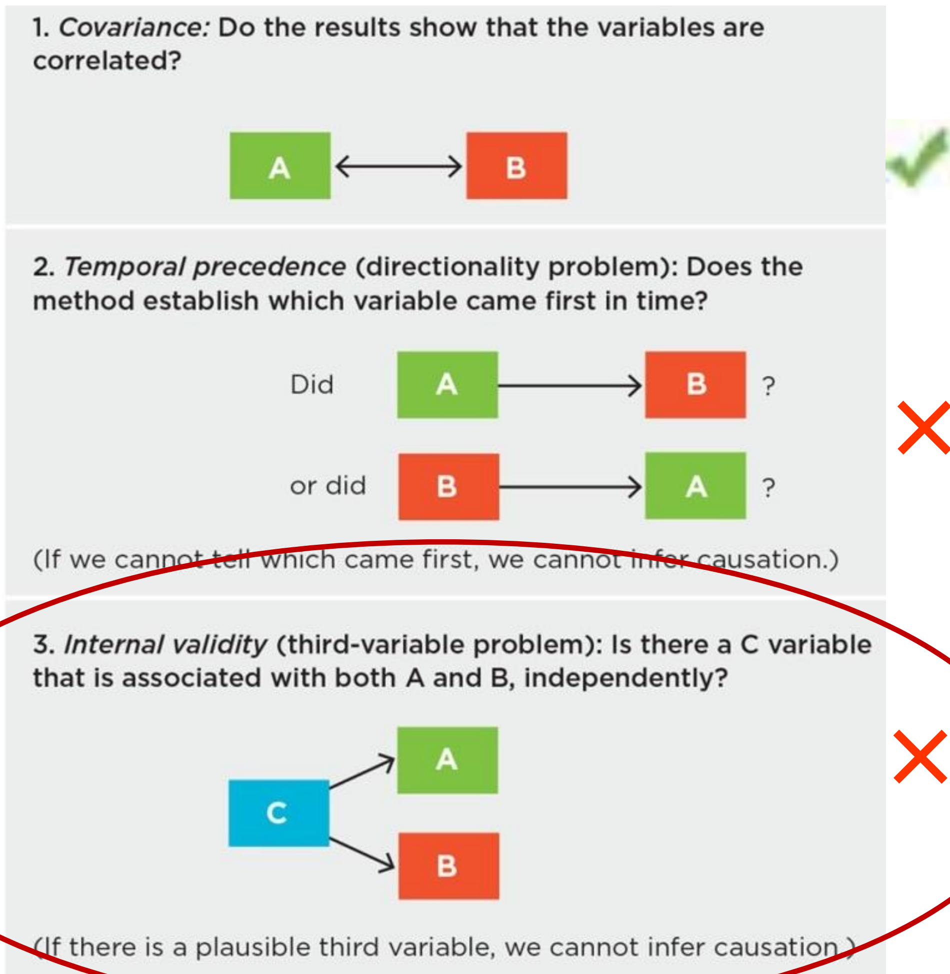
# Correlational research

- Correlational research → association claims
  - The presence of a correlation indicates an association but does not indicate causality!
    - directionality problem
    - 3rd variable problem/ Spurious correlations

# Correlational research

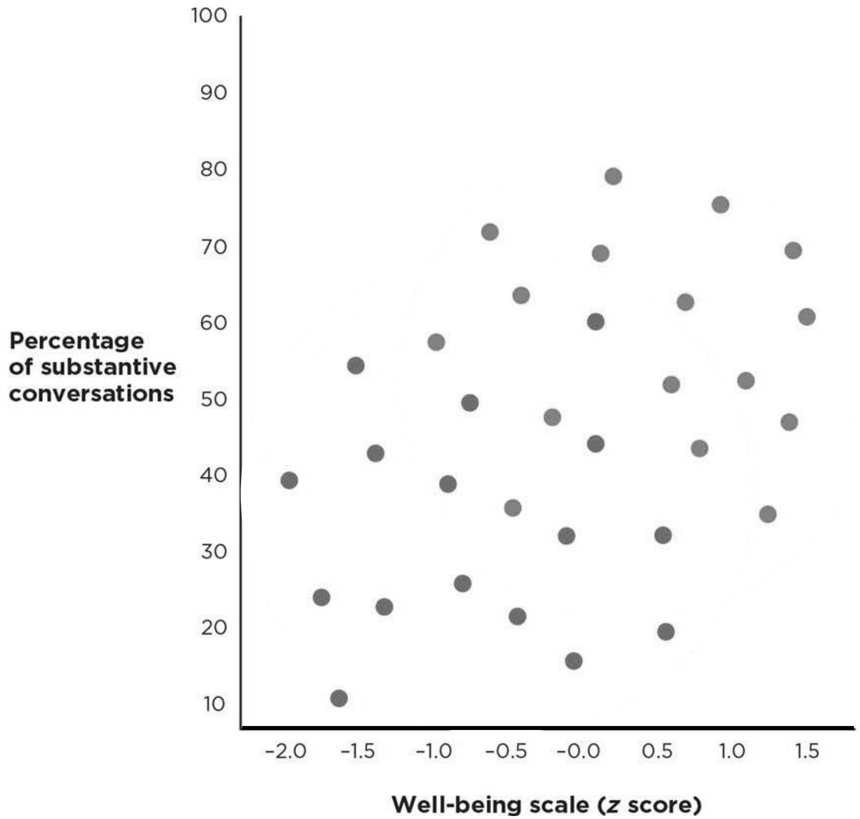
- What are the criteria required to establish causation?




- ***3rd variable problem*** Sometimes researchers attempt to establish (some) internal validity based on correlational research
  → How?

# Correlational research

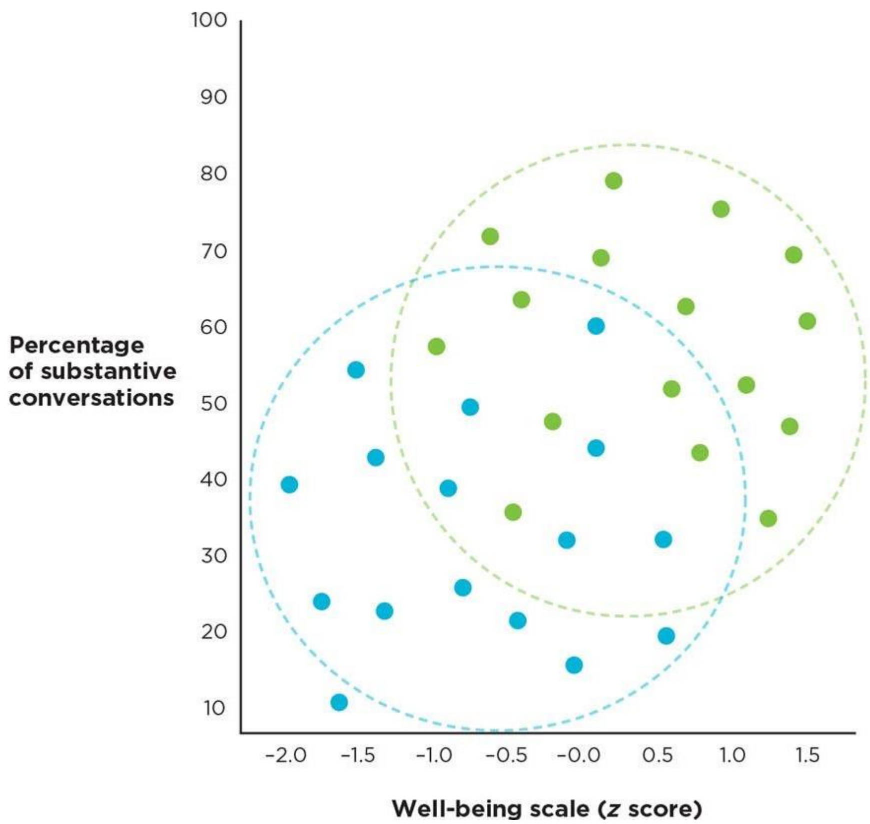
- By separating the participants into subgroups based on a reasonable 3rd variable




Overall, there is a moderate positive correlation.

# Correlational research

- by separating the participants into subgroups on the basis of a reasonable 3rd variable

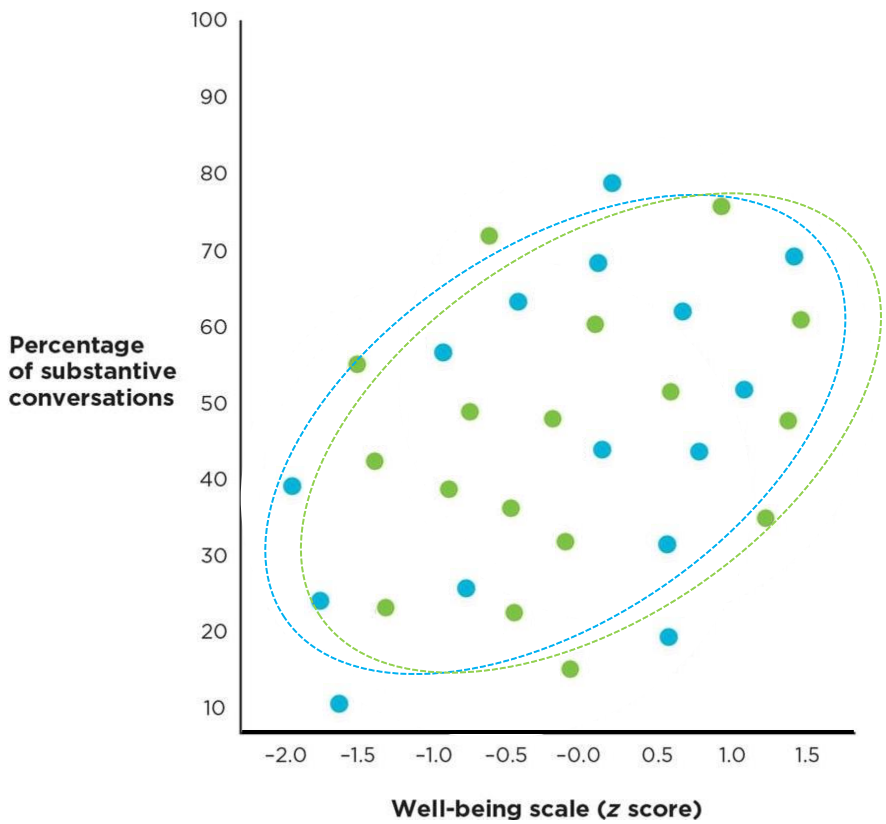



Overall, there is a moderate positive correlation. Within the two subgroups there is no positive correlation.

→ Education **does** present a 3rd variable problem

# Correlational research

- by separating the participants into subgroups on the basis of a reasonable 3rd variable



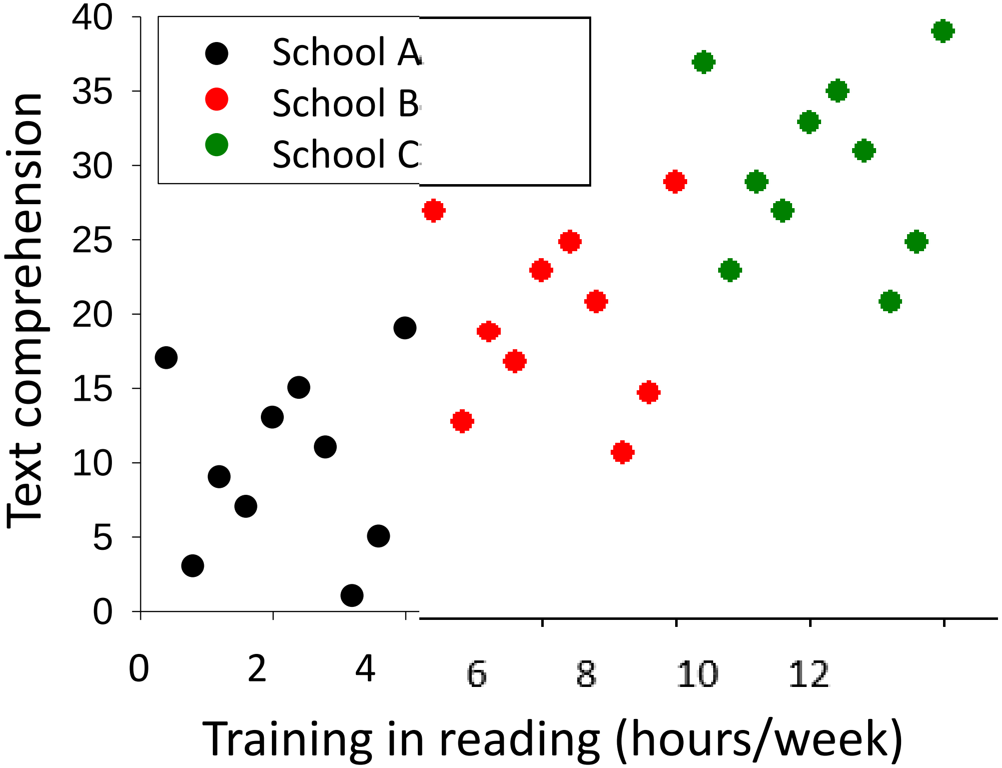

Overall, there is a moderate positive correlation. Within the two subgroups there is also a positive correlation.

→ Education **does <u>not</u>** present a 3rd variable problem

# Correlational research

- Correlational research → association claims
  - The presence of a correlation indicates an association but does not indicate causality!
  - What if the correlation is zero?
    - Restriction of range (or truncated range)

# Correlational research

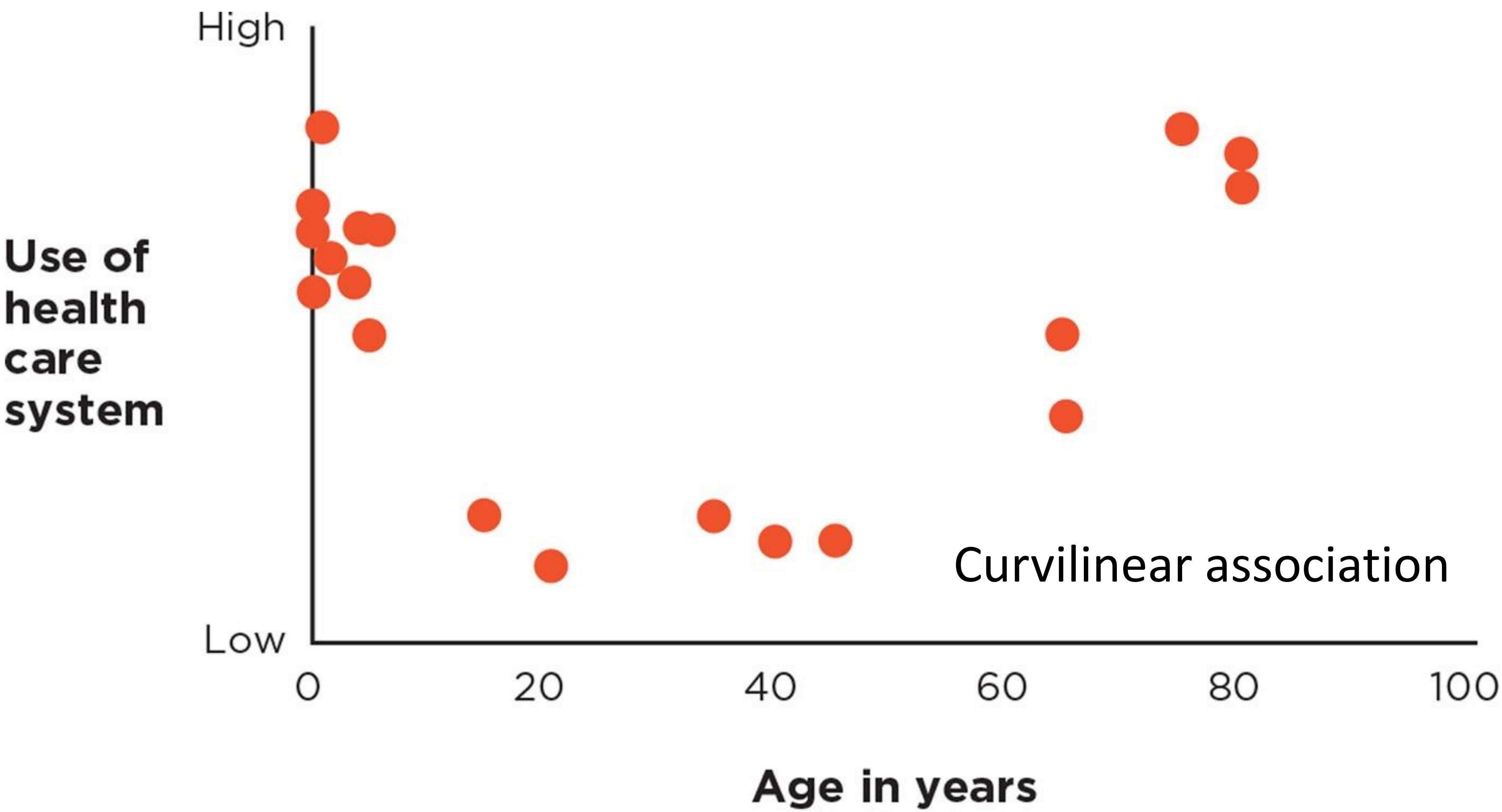



Per school: $r$ = .02;
Across schools: $r$ = .78

# Correlational research

- Correlational research → association claims
  - The presence of a correlation indicates an association but does not indicate causality!
  - What if the correlation is zero?
    - Restriction of range (or truncated range)
    - Non-linear relationship

# Correlational research




Curvilinear association

# Correlational research

- Correlational research → association claims
  - The presence of a correlation indicates an association but does not indicate causality!
  - What if the correlation is zero?
    - Restriction of range (or truncated range)
    - Non-linear relationship
    - No association

# Correlational research

- Correlational research → association claims
  - The presence of a correlation indicates an association but does not indicate causality!
  - The internal validity of associations claims is low
  - It is important to question how well
    - each variable was measured  (construct validity)
    - the results generalize (external validity)
    - the numbers support the conclusions (statistical validity)

# Correlational research

A researcher is interested in the relationship between degree of religiosity and attitude on pornography.

She performs a study in Hardinxveld, a religious village in the Alblasserwaard.

By using long questionnaires, she obtains a reliable indication of the degree of religiosity and the attitude on pornography of 75% of the inhabitants.

The results showed a zero correlation ($r = -.03$). She concludes on the basis of her result that religion is unrelated to attitude to pornography.

Please indicate to what extent you agree/disagree with this conclusion.

# Correlational research

The New York Times

PHYS ED

## Even a Little Exercise Might Make Us Happier ?

iStock

By Gretchen Reynolds

May 2, 2018

查看简体中文版 - 查看繁體中文版

Small amounts of exercise could have an outsize effect on happiness.

According to a new review of research about good moods and physical activity, people who work out even once a week or for as little as 10 minutes a day tend to be more cheerful than those who never exercise. And any type of exercise may be helpful.

The idea that moving can affect our moods is not new. Many of us would probably say that we feel less cranky or more relaxed after a jog or visit to the gym.

Science would generally agree with us. A number of past studies have noted that physically active people have much lower risks of developing depression and anxiety than people who rarely move.

108

https://www.nytimes.com/2018/05/02/well/move/even-a-little-exercise-might-make-us-happier.html

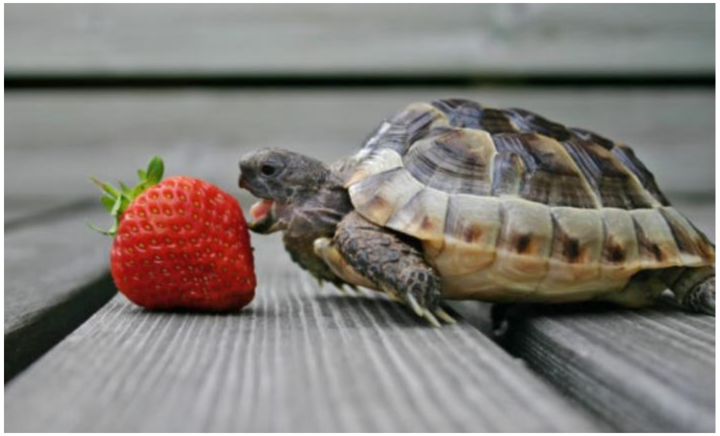
# Correlational research

?

Are You a Fast Eater? Slow Down to Eat (and Weigh) Less

Let your brain catch up with your stomach

SHARE



Mom was right when she told us to slow down when we ate. Mom promised that wolfing down your food would likely result in a stomachache, but a new study shows something of more consequence: Eating slower could be linked to a lower risk of obesity.

Researchers from a university in Japan examined data from 59,717 people with type 2 diabetes. Researchers asked people to describe themselves as fast eaters, medium eaters or slow eaters.

“People who were slowest eaters had the lowest risk of obesity. People who self-described as medium-eaters had a bit higher risk, but the highest risk was in the fast-eating group,” says psychologist Leslie Heinberg, PhD, MA.

May 29, 2018

https://health.clevelandclinic.org/are-you-a-fast-eater-slow-down-to-eat-and-weigh-less/

# The basics of experimenting

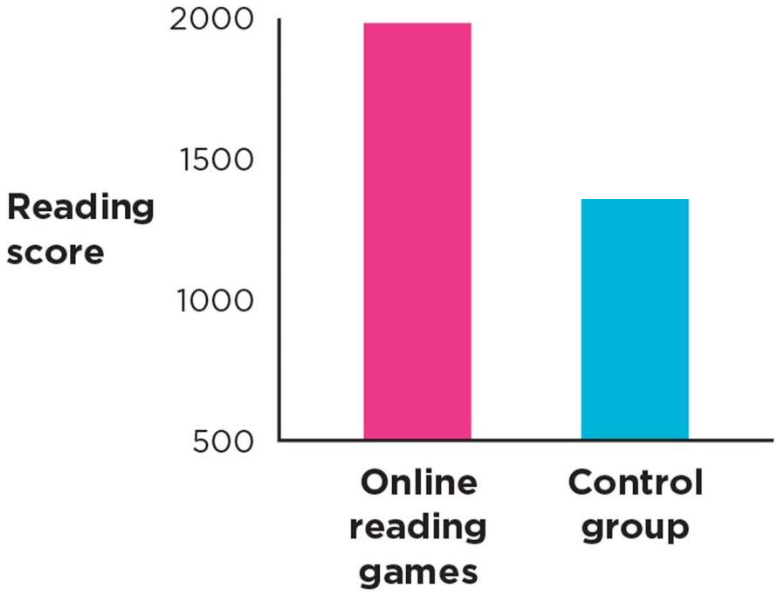
- An experiment occurs when a particular comparison is produced while other aspects of the situation are held constant.

# The basics of experimenting

- A hypothetical experiment
  - Do online reading games make kids better readers?
    - Independent variable:
      - Reading training
        - online reading games (experimental group)
        - no online reading games (control group)
    - Dependent variable:
      - Reading ability
        - operationalized in terms of score on reading test

# The basics of experimenting

- A hypothetical experiment
  - Do online reading games make kids better readers?




Note: these results are fabricated!

# The basics of experimenting

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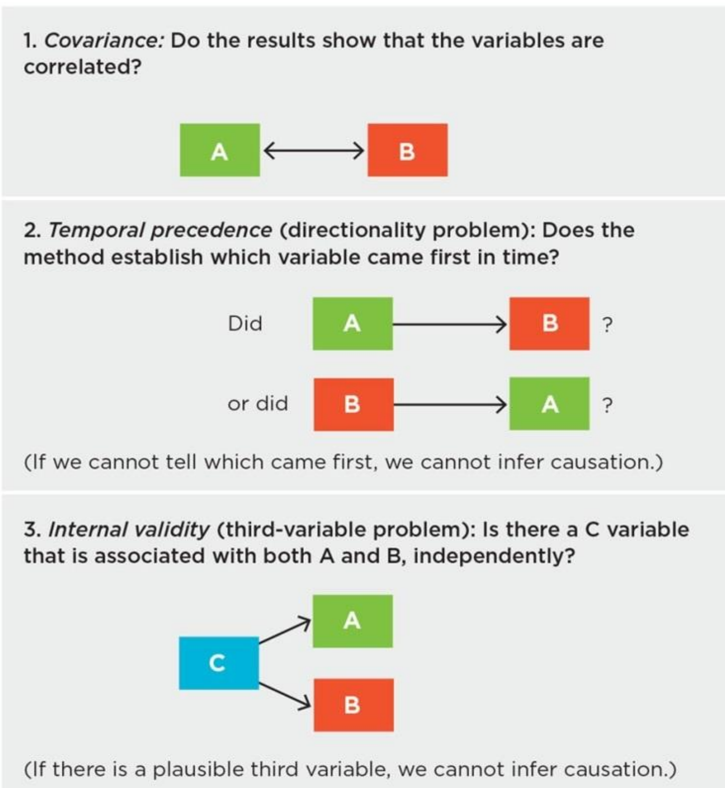


# The basics of experimenting

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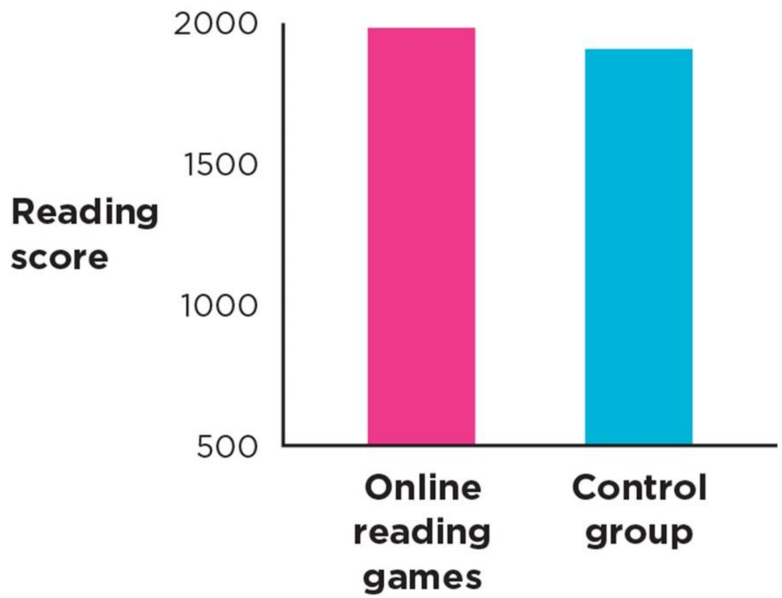
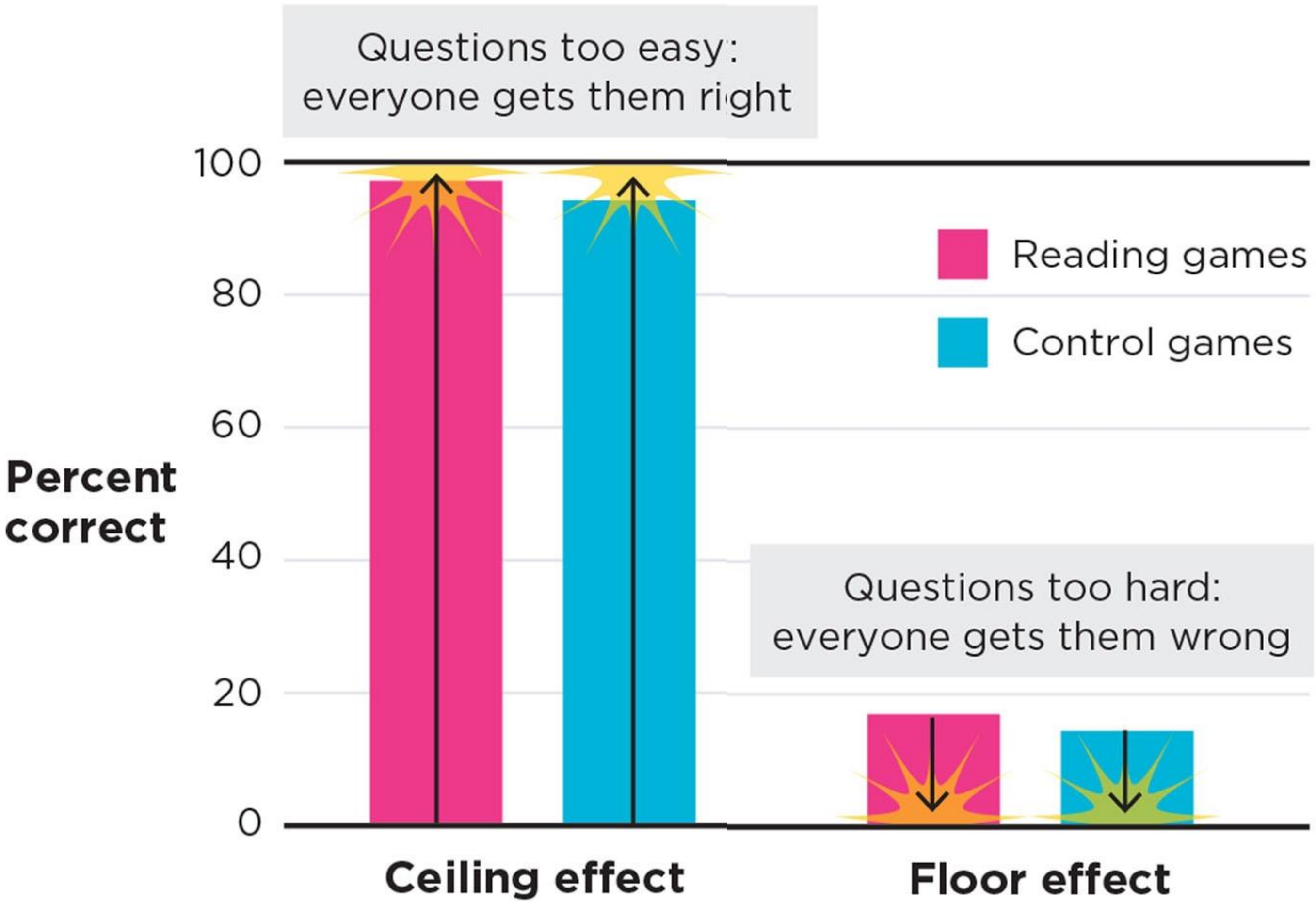



Note: these results are fabricated!

# Experimental research

- Null results ($H_0$)
  - The independent variable did not affect the dependent variable
  - Manipulation was too weak
    - e.g., 1 week of training might have been too short
  - Insensitive measures
    - e.g., tiny improvements in reading ability might not have been detectable in a simple reading test
  - Ceiling effect
    - e.g., reading test might have been too easy
  - Floor effect
    - e.g., the reading test might have been too difficult.

# Experimental research



# Experimental research

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    - Ceiling effect
        - e.g., reading test might have been too easy
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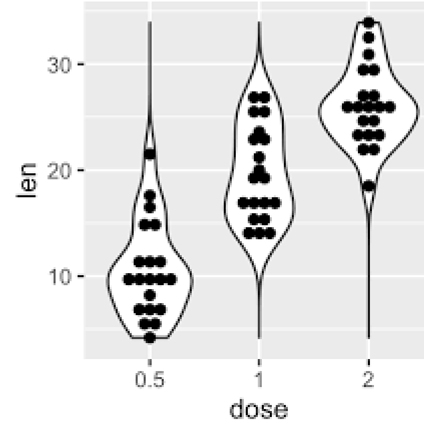
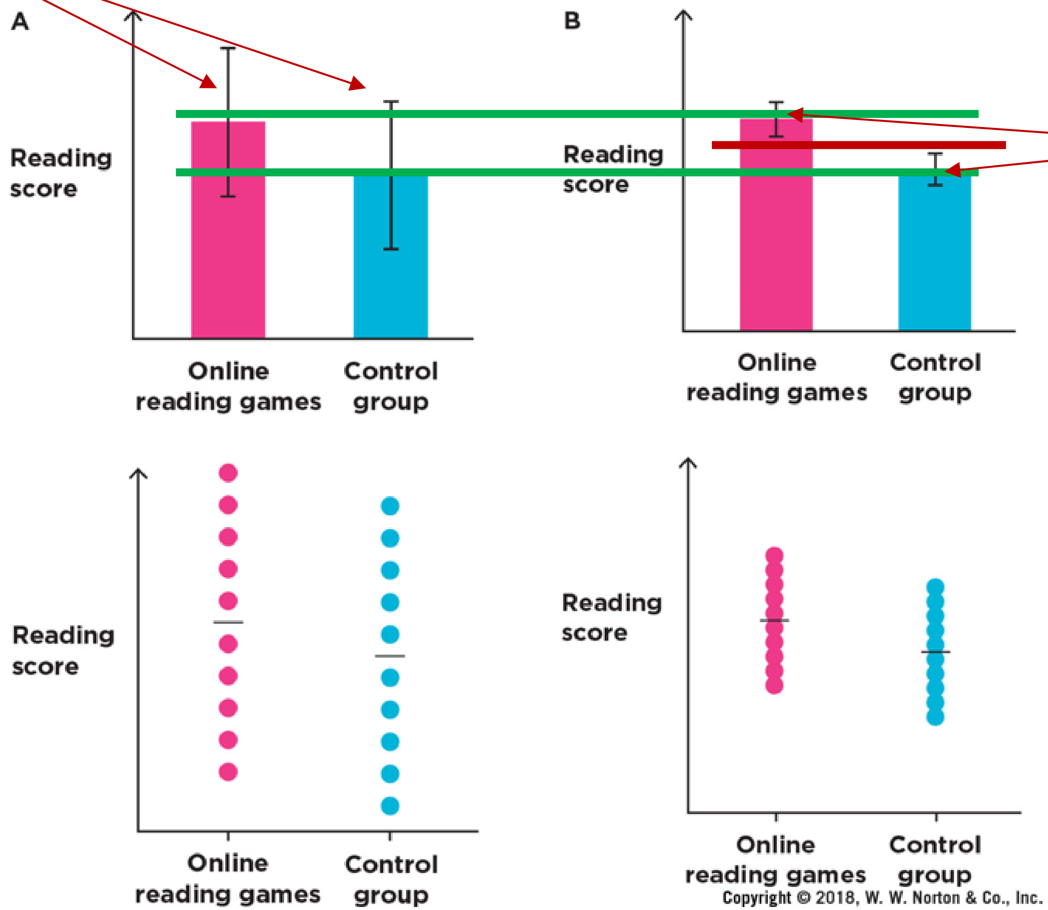
        - e.g., reading test might have been too difficult
    - Unreliable measurements

# Experimental researc

Error bars




Error bars

# Experimental research

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    - e.g., reading test might have been too easy
  - Floor effect
    - e.g., reading test might have been too difficult
  - Unreliable measurements
  - Not enough power (small n)

# Experimental research

A scientist is interested in the possible effects of caffeine on performance in a memory task.
To investigate this, she conducts an experiment with two groups.
The participants in one group receive a cup of regular coffee whereas the participants in the other group receive a cup of decaffeinated coffee.
Both groups take subsequently part in a memory task.
The results show no difference in task performance between both groups.

Could you think of possible explanations for this result?

# Experimental research

- Experiment (Brady, Porter, Conrad, & Mason, 1958)

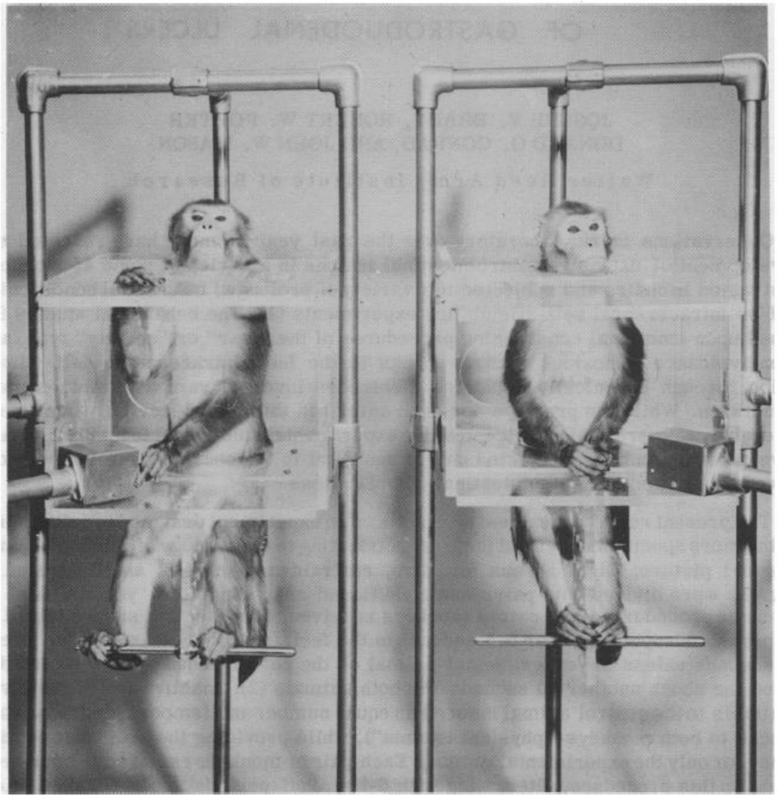


Fig. 1. An experimental monkey and a control monkey, gently restrained in primate chairs, illustrate the "yoked-chair" avoidance situation. The lever available to each animal is shown within easy reach, although only the experimental "avoidance" monkey on the left is observed to press the lever.

# Experimental research

- Experiment (Jarrard, 1963)

Psychopharmacologia 5, 39—46 (1963)

From the Washington and Lee University

**Effects of D-Lysergic Acid Diethylamide on Operant Behavior in the Rat***

By

LEONARD E. JARRARD

With 3 Figures in the Text

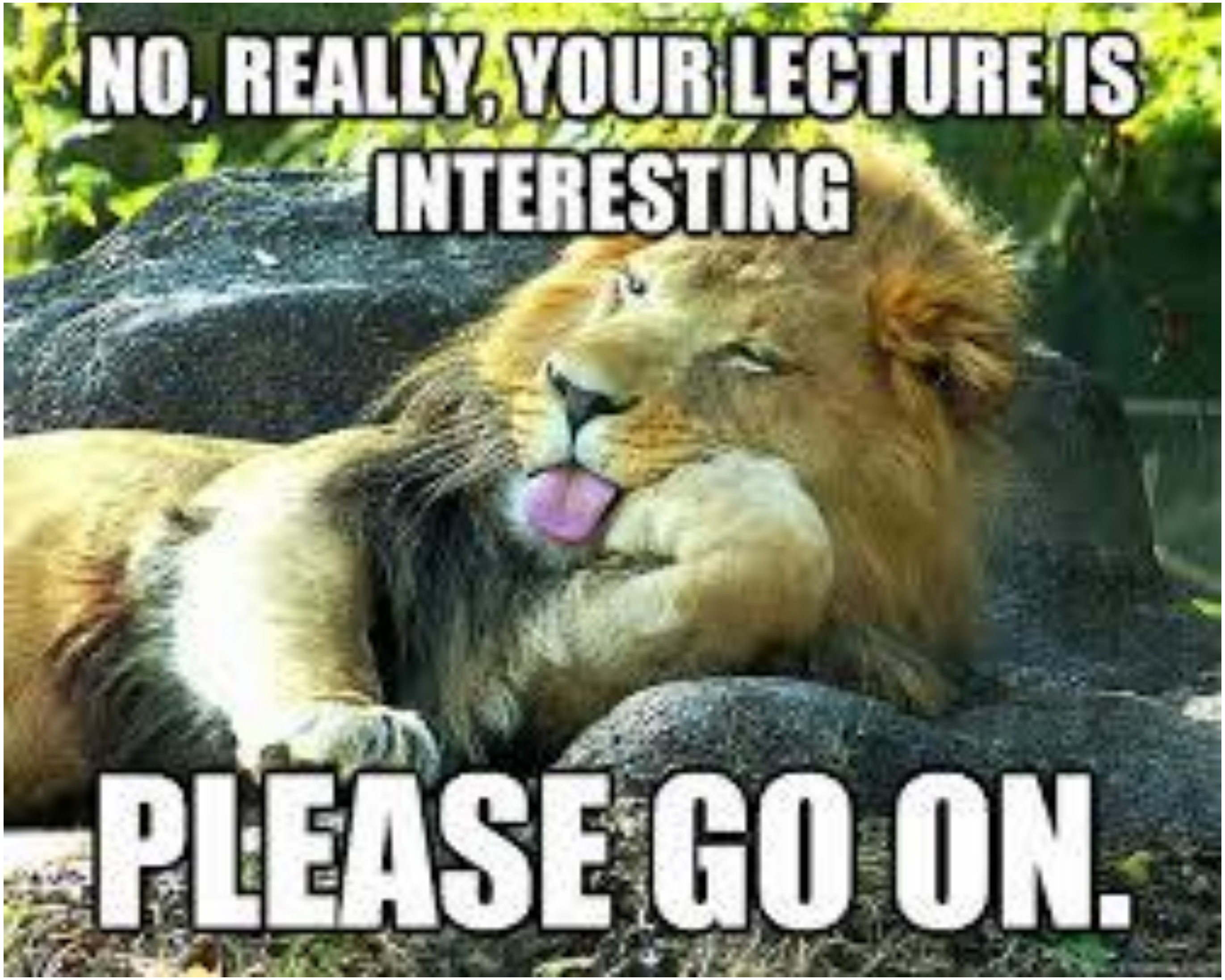
*(Received January 30, 1963)*

# Experimental research

- How to assign subjects to the different levels of the independent variable(s)?
  - **Between-subjects design** (independent-groups design)
    - Separate groups of subjects receive different levels of the independent variable(s).
  - **Within-subjects design** (within-groups/repeated-measures design)
    - All subjects receive all levels of the independent variable(s).

# Within & Between Subjects Designs

| **Independent variable** | **Dependent variable** |
|---|---|
| Location of lesion in the cortex | % correct |
| Rotation speed of radar arm | % detected |
| Learning method | Performance on test |
| Odor | Score concerning the attractiveness of person |


NO, REALLY, YOUR LECTURE IS INTERESTING
PLEASE GO ON.